

ECO_LOESS_Filter

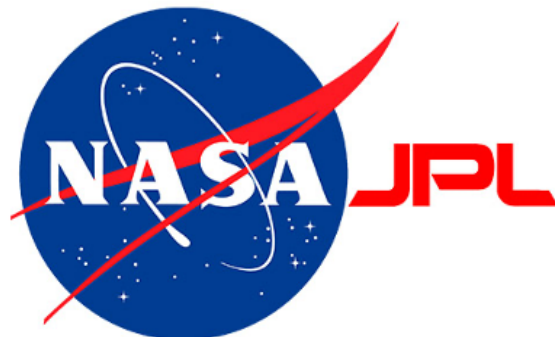
NASA JPL ECOSTRESS Land Surface Temperature LOESS Outlier Removal



User Guide & Reference Manual

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1 Introduction

The ECOSTRESS Land Surface Temperature (LST) Locally Estimated Scatterplot Smoothing (LOESS) Graphical User Interface (GUI) is an interactive Python application for tuning and applying LOESS-based outlier removal to ECOSTRESS LST GeoTIFF scenes. It is designed to remove cloud fringes, isolated anomalously cold pixels, and other hot or cold outliers that may remain after earlier preprocessing steps such as QC filtering, geolocation correction, and land masking.

This guide describes how to use LOESS_v16.py to build LOESS baseline graphs, preview the effect of filtering parameters on representative scenes, save tuning snapshots, and apply the final LOESS filter across all detected LST GeoTIFFs.

2 LOESS Graphical User Interface (GUI) Overview

The LOESS GUI recursively scans a selected working folder for LST GeoTIFF scenes, builds a LOESS baseline from randomly sampled calibration pixels in the shared overlap region, and applies hot/cold outlier thresholds relative to that baseline.

Specifically, the GUI:

- Scans a selected folder for ECOSTRESS LST GeoTIFF files.
- Builds diagnostic baseline graphs from selected calibration pixels.
- Displays individual pixel baselines and the averaged LOESS baseline for comparison.
- Previews an original scene beside the LOESS-filtered version.
- Displays the selected sample pixels as points on the scene preview.
- Allows users to adjust LOESS parameters, color ramp settings, sample pixel color, and viewer behavior.
- Saves tuning snapshots including graphs, preview TIFFs, parameters, and sample pixels.
- Applies the current LOESS filter to all detected scenes into a LOESS_filtered folder.
- Optionally reprojects LOESS outputs to EPSG:4326 after processing if not already.

3 Setup Requirements

Software & Libraries

Recommended Python Version: 3.10 - 3.12, 64-bit. The script may run on nearby versions, but the recommended range is safest for geospatial libraries.

Install the following Python libraries:

- numpy
- rasterio
- affine
- statsmodels
- PyQt5

- matplotlib
- fiona (needed for shapefile reprojection only)

Install command: If packages are missing, install them in Terminal using: `pip install numpy rasterio affine statsmodels PyQt5 matplotlib fiona`

Input Files

Place LOESS_v16.py in the folder that contains your ECOSTRESS LST GeoTIFF scenes, or launch the GUI and browse to the folder. The selected folder may contain subfolders; the GUI searches recursively.

- ECOSTRESS LST GeoTIFF files, typically filenames containing LST.
- All scenes should share the same CRS and have a shared overlapping area.
- At least 3 scenes are required to build a LOESS time series, but more will produce better results (~200 is recommended).



Important: The GUI does not process files already inside LOESS_filtered or LOESS_tuning_parameters during the main scan, which prevents re-ingesting previous outputs.

4 Launching the GUI

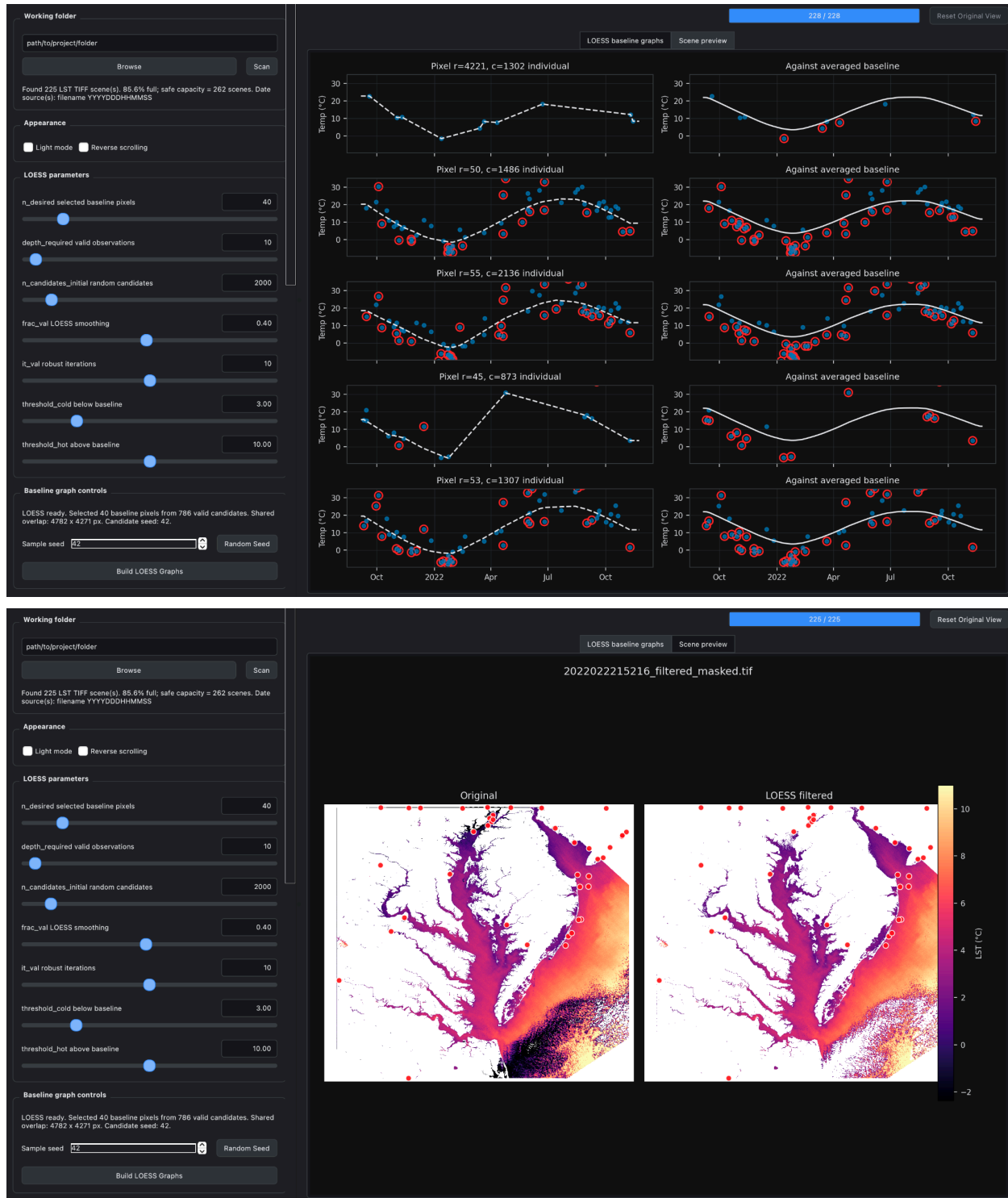
Step 1: Open Terminal or your preferred Python environment.

Step 2. Navigate to the folder containing LOESS_v16.py.

Operating System	Example Command
macOS / Linux	<code>cd /path/to/project/folder</code>
Windows	<code>cd C:\path\to\project\folder</code>

Step 3. Run the script: `python LOESS_v16.py`

Tip: On some systems the command may be `python3 LOESS_v16.py` instead of `python LOESS_v16.py`.



GUI visual after launching the script, building LOESS graphs, and previewing an example scene. This example shows a scene from Chesapeake Bay, Maryland and Virginia. The left panel contains scene level controls and the right panel displays the scene preview.

5 GUI Panel Reference

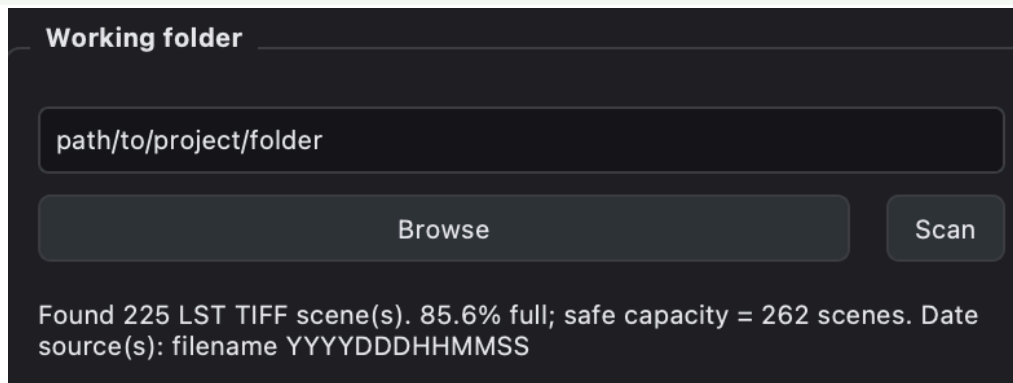
The GUI is organized into a left control menu and a large right viewer panel. The left panel contains all settings; the right panel contains two tabs: LOESS baseline graphs and Scene preview.

5.1 Working Folder Control

The Folder control sets the root folder used for scanning. The Browse button opens a folder picker. The Scan button searches the selected folder recursively for LST GeoTIFF scenes.

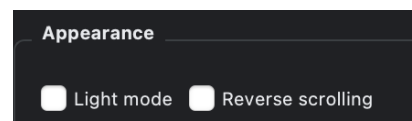
After scanning, the panel reports how many LST TIFF scenes were found, the percent full capacity estimate, the estimated safe scene capacity, and the date source used for ordering scenes.

Scan behavior: The GUI excludes LOESS output, tuning, and reprojection folders from scans to avoid accidentally processing previous outputs again.

A screenshot of the 'Working folder' GUI panel. It features a dark background with a light-colored text input field containing the placeholder text 'path/to/project/folder'. Below the input field are two buttons: 'Browse' and 'Scan'. At the bottom of the panel, there is a status message: 'Found 225 LST TIFF scene(s). 85.6% full; safe capacity = 262 scenes. Date source(s): filename YYYYDDHMMSS'.

5.2 Appearance Control

The Appearance parameters control interface behavior and display preferences.

A screenshot of the 'Appearance' GUI panel. It has a dark background and contains two toggle switches. The first switch is labeled 'Light mode' and is currently turned off. The second switch is labeled 'Reverse scrolling' and is also currently turned off.

Control	Use
Light mode	Switches the interface between dark and light themes.
Reverse scrolling	Reverses the mouse wheel or trackpad zoom direction in the viewer.

5.3 LOESS Parameters

These parameters control how the LOESS baseline is built and how hot/cold outliers are detected. Each parameter has a hover tooltip in the GUI. The numeric value boxes are editable: click the number, type an exact value, and press Enter or click away.

Parameter	Default	Use
n_desired	40	Number of pixels used to build the LOESS averaged baseline.
depth_required	10	Minimum number of valid observations required per candidate pixel.
n_candidates_initial	2000	Number of random pixels sampled initially when searching for calibration candidates.
frac_val	0.40	LOESS smoothing fraction. Higher values create a smoother baseline; lower values allow the baseline to follow data more closely but can look noisy.
it_val	10	LOESS robustness iterations used to downweight outliers during smoothing.
threshold_cold	3.00	Outlier threshold below the baseline, in degrees Celsius.
threshold_hot	10.00	Outlier threshold above the baseline, in degrees Celsius.

Parameter guidance: If too few valid calibration pixels are found, increase n_candidates_initial or lower depth_required. If the baseline is too noisy, increase frac_val. If the baseline is too smooth, lower frac_val. If too many pixels are removed, increase threshold_cold or threshold_hot. If cloud fringes remain, make thresholds stricter and inspect the preview.

The image shows a dark-themed user interface for adjusting LOESS parameters. It features eight rows, each with a parameter name, a descriptive label, a slider control, and a numeric input box. The parameters and their current values are: n_desired (40), depth_required (10), n_candidates_initial (2000), frac_val (0.40), it_val (10), threshold_cold (3.00), and threshold_hot (10.00). The sliders are represented by blue circular handles on horizontal tracks.

LOESS parameters	
n_desired selected baseline pixels	40
depth_required valid observations	10
n_candidates_initial random candidates	2000
frac_val LOESS smoothing	0.40
it_val robust iterations	10
threshold_cold below baseline	3.00
threshold_hot above baseline	10.00

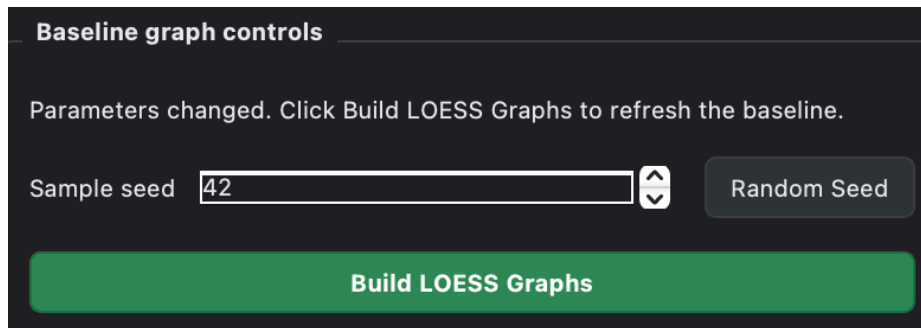
5.4 Baseline Graph Controls

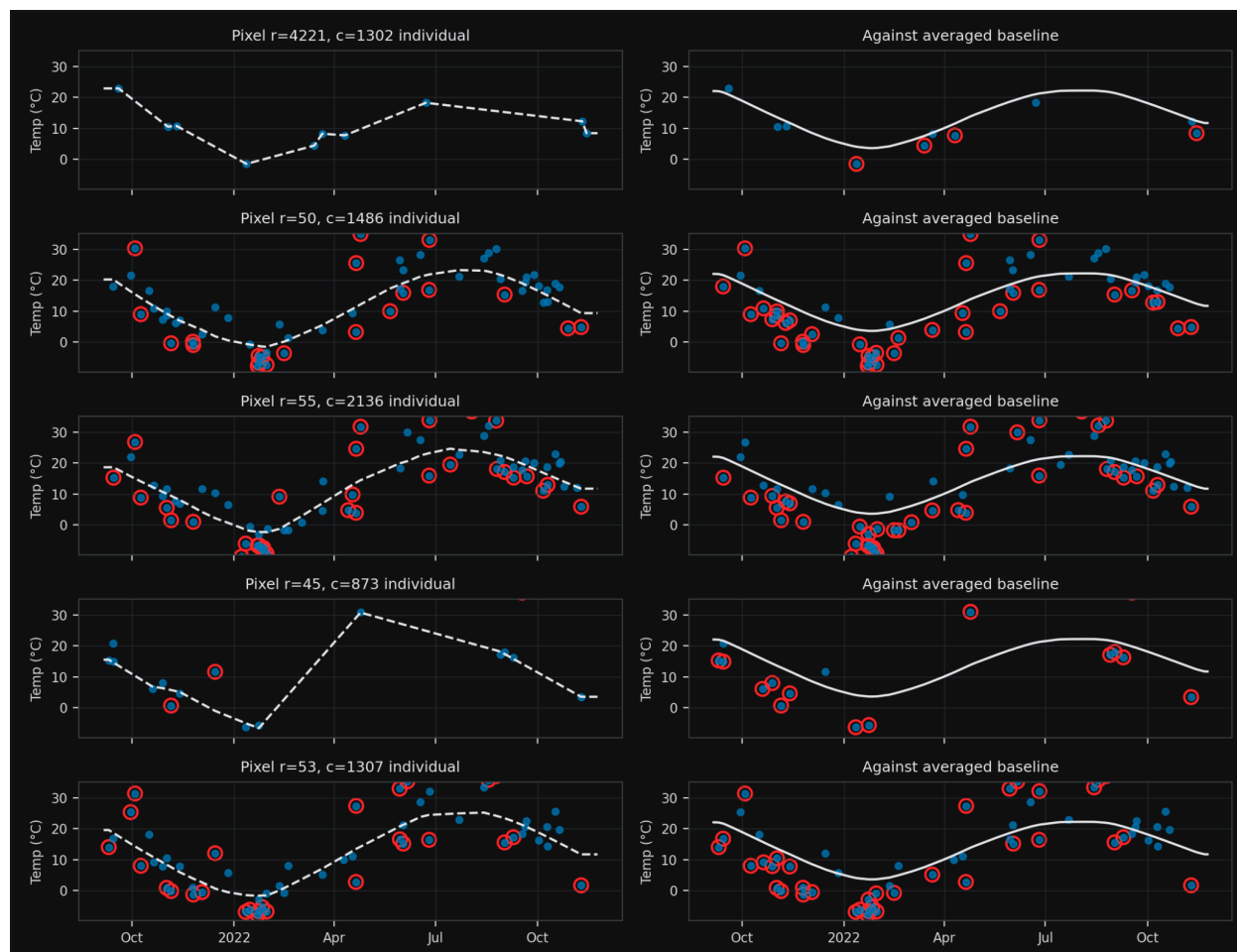
The Baseline graph controls build the LOESS model. The Build LOESS Graphs button turns green when the graphs need to be rebuilt, including when the GUI first opens or when a parameter changes.

Control	Use
Sample seed	Controls the random candidate-pixel sample. Reusing the same seed recreates the same candidate sample.
Random Seed	Generates a new random seed for a different candidate-pixel sample.
Build LOESS Graphs	Builds the LOESS baseline using the current folder, seed, and parameter settings.

After clicking Build LOESS Graphs, a confirmation popup summarizes the number of scenes, sample seed, and estimated graph generation time. The top progress bar updates while the baseline is being built.

The LOESS baseline graphs tab shows selected calibration pixels. The left column compares each selected pixel with its individual LOESS baseline, shown as a dashed line. The right column compares that same pixel with the averaged LOESS baseline, shown as a solid line.



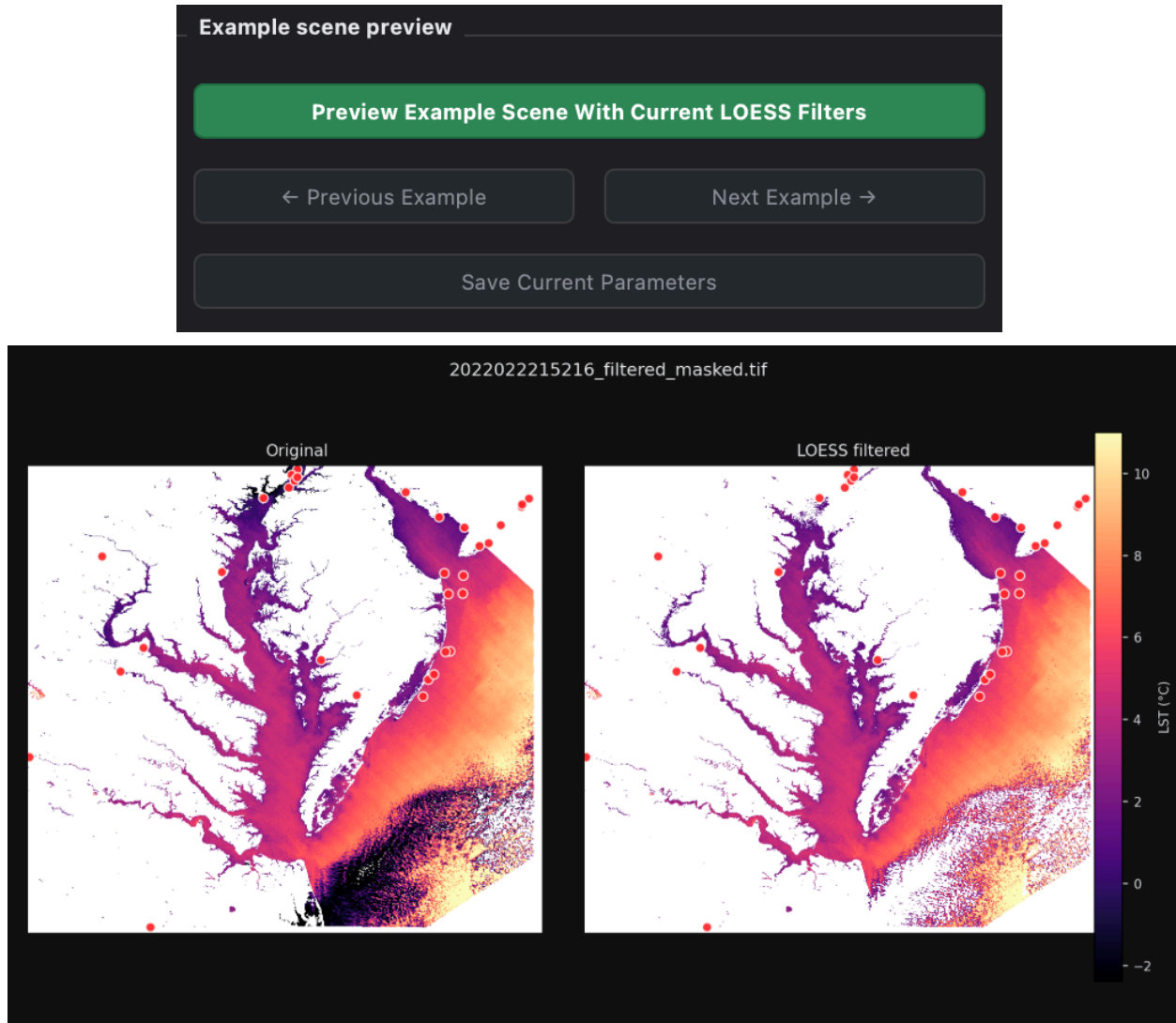


5.5 Example Scene Preview

After the LOESS graphs are built, the Preview Example Scene With Current LOESS Filters button turns green. Click it to select and display a representative scene using the current LOESS filter.

Control	Use
Preview Example Scene With Current LOESS Filters	Builds the side-by-side scene preview using the current LOESS model.
Previous Example / Next Example	Cycles through example scenes.
Save Current Parameters	Saves graphs, preview TIFFs, parameters, and sample pixels for the current tuning run.

The scene preview displays two synchronized panels: Original and LOESS filtered. The same zoom and pan view is applied to both panels. The selected sample pixels used to build the averaged baseline are plotted as points on both panels.



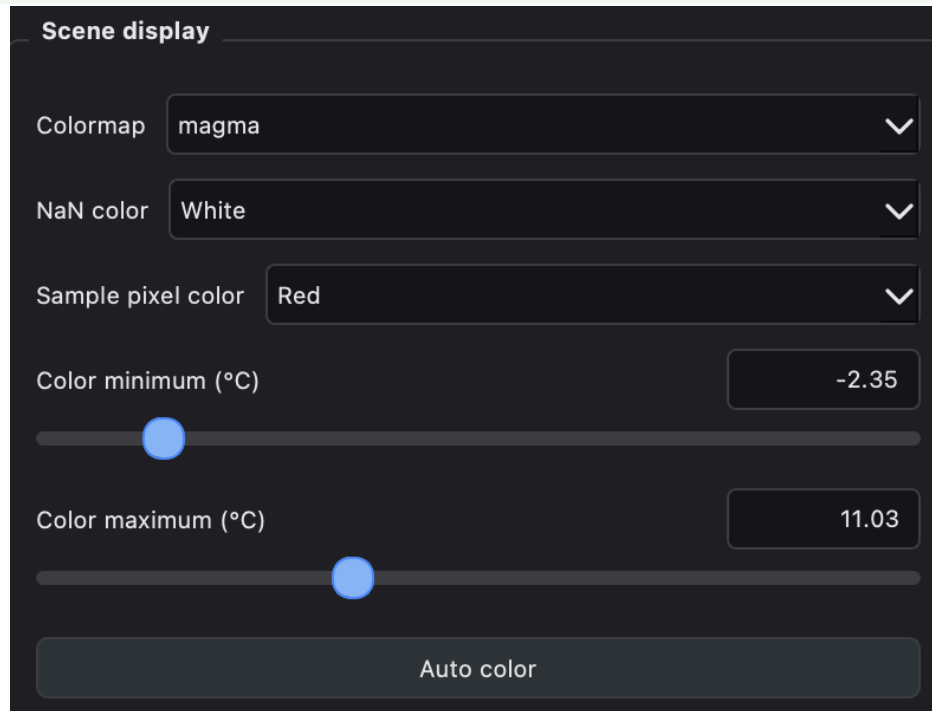
5.6 Scene Display

The Display control affects only how the preview appears. These settings do not change the saved raster values. They help users visually decide whether the LOESS parameters are appropriate.

Control	Use
Colormap	Changes the preview color ramp.
NaN color	Changes how masked or missing pixels are displayed in the preview.
Sample pixel color	Changes the color of the calibration sample points plotted on the scene preview.

Color minimum / maximum	Controls the Celsius display stretch. These values are editable.
Auto color	Sets the color stretch from the current preview.

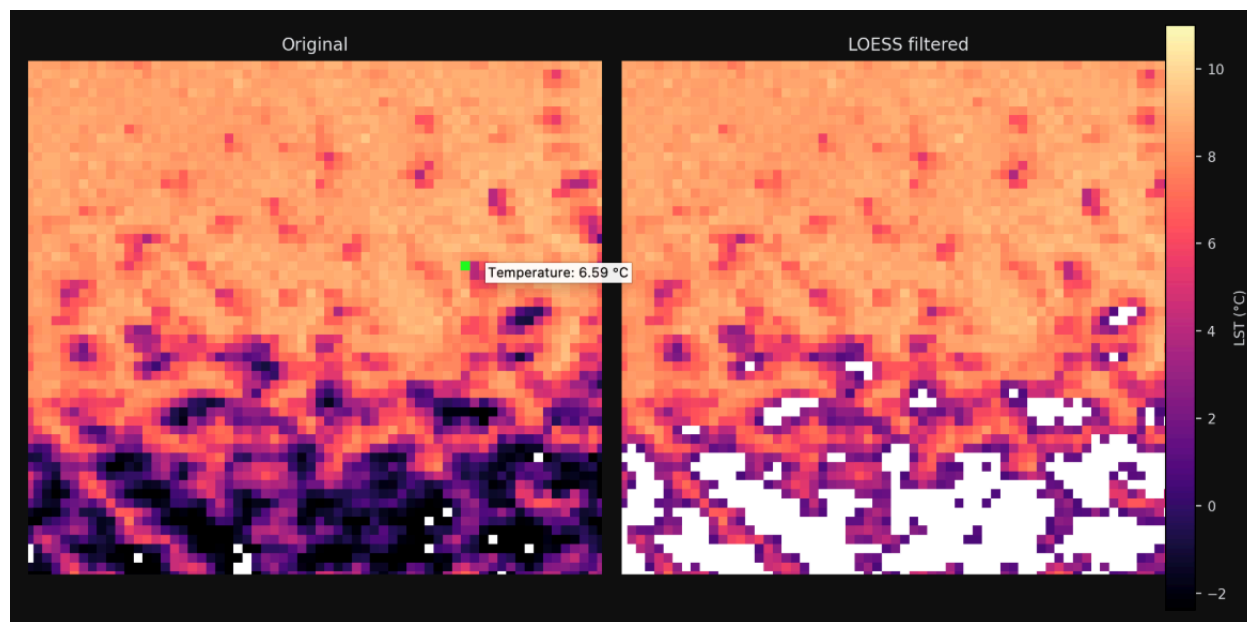
Display note: The color minimum and maximum auto-generate when previewing or switching example scenes. Manually edited color values are useful for visual inspection but do not change the output data.



5.7 Viewer Control

The viewer shows the selected representative scene with the current mask applied.

Action	Result
Mouse wheel / trackpad scroll	Zooms in or out. Direction can be reversed from the Appearance panel.
Left-click and drag	Pans around the scene.
Single left-click on a pixel	Shows a popup with temperature in °C.
Right-click over the viewer / Reset Original View button	Resets to the original full-scene view.



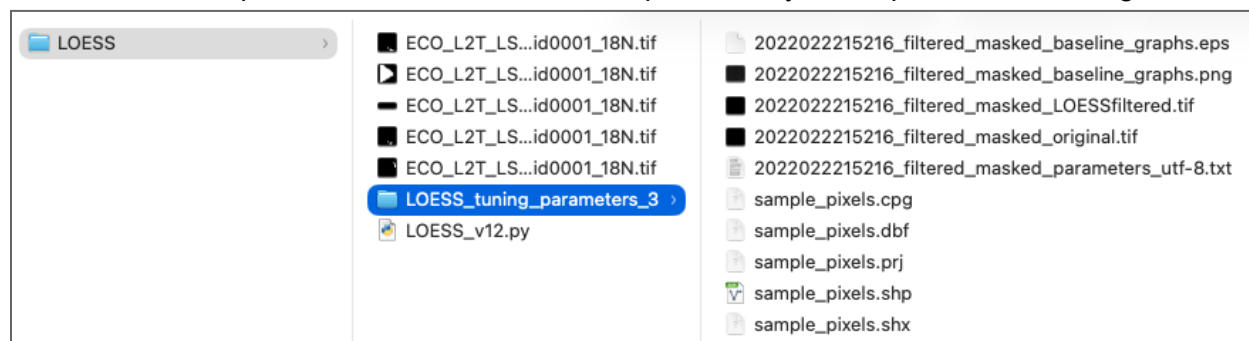
5.8 Save Current Parameters

Save Current Parameters creates a new tuning snapshot folder each time it is clicked. The first save goes to LOESS_tuning_parameters. Later saves go to LOESS_tuning_parameters_1, LOESS_tuning_parameters_2, and so on.

In the saved folder:

- <scene_stem>_baseline_graphs.png
- <scene_stem>_baseline_graphs.eps
- <scene_stem>_original.tif
- <scene_stem>_LOESSfiltered.tif
- <scene_stem>_parameters_utf-8.txt
- sample_pixels.shp and its shapefile files

The parameter text file records the LOESS settings, seed, selected baseline pixel count, shared overlap dimensions, display settings, and output paths. The sample_pixels.shp file stores the selected baseline pixel locations so users can inspect exactly which pixels were averaged.



5.9 Apply to All Control

The Apply to all scenes control writes LOESS-filtered GeoTIFF outputs for every scene in the current LOESS model.

Option	Behavior
Append copies into LOESS_filtered with suffix	Keeps source scenes unchanged and writes new suffixed copies into LOESS_filtered. Default suffix is _LOESS_filtered.
Overwrite files inside LOESS_filtered using original filenames	Writes outputs into LOESS_filtered using original filenames. Source input scenes are not modified.
Apply Current LOESS Filter to All LST TIFFs	Applies the current built LOESS model across all detected scenes.
Reproject LOESS Outputs to EPSG:4326	Scans LOESS_filtered for non-EPSG:4326 outputs and opens the reprojection dialog.

Apply to all scenes

☒ Append copies into LOESS_filtered with suffix (keep source scenes)
☐ Overwrite files inside LOESS_filtered using original filenames

Append suffix

All LOESS outputs are written to LOESS_filtered. Source input scenes are not modified. Use the EPSG:4326 option after the batch if the outputs need reprojection.

6 Output Files and Folder Structure

The GUI writes outputs into fixed folders inside the selected project folder.

Output Location	Contains
LOESS_filtered	LOESS-filtered GeoTIFF scenes produced by Apply Current LOESS Filter to All LST TIFFs.
LOESS_filtered/reprojected_EPSG4326	Appended EPSG:4326 copies of LOESS outputs, if selected in the reprojection dialog.

LOESS_tuning_parameters	First saved parameter snapshot folder.
LOESS_tuning_parameters_1, _2, ...	Additional saved parameter snapshot folders created by later Save Current Parameters runs.

Mode	LOESS Output
Append	New file in LOESS_filtered with _LOESS_filtered suffix. Source scene stays in the input folder.
Overwrite	File goes to LOESS_filtered using the original filename. Source scene stays in the input folder.

Recommended practice: Use append mode while tuning or testing so source scenes and earlier outputs remain untouched. Use overwrite mode only when you are intentionally replacing existing LOESS_filtered products.

7 EPSG:4326 Reprojection

After the batch LOESS run is complete, the GUI can scan LOESS_filtered for non-EPSG:4326 TIFF files. Original input-folder scenes are not queued for reprojection.

Reprojection Mode	Behavior
Append converted files into reprojected_EPSG4326 with suffix	Keeps original LOESS outputs and writes EPSG:4326 copies into reprojected_EPSG4326 folders. Default suffix is _EPSG4326.
Overwrite LOESS outputs in place	Replaces queued LOESS_filtered outputs with EPSG:4326 versions.

Scope: The reprojection scanner skips existing reprojected_EPSG4326 folders so appended reprojection outputs are not processed again.

Reproject LOESS outputs to EPSG:4326

Only files inside LOESS_filtered are queued. Append mode keeps the LOESS outputs and writes converted copies into reprojected_EPSG4326 with the EPSG4326 suffix.

116 LOESS output TIFF file(s) will be converted to EPSG:4326.

Folder:

Output mode

- ☒ Append converted files into reprojected_EPSG4326 with suffix (keep original files)
- ☐ Overwrite existing files in place

Suffix

Ready to reproject.

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Files queued

[PENDING] LOESS_filtered/ECO_L2T_LSTE.002_LST_doy2020060005418_aid0001_2S_LOESS_filtered.tif (EPSG:32702 to EPSG:4326)
[PENDING] LOESS_filtered/ECO_L2T_LSTE.002_LST_doy2020063100150_aid0001_2S_LOESS_filtered.tif (EPSG:32702 to EPSG:4326)
[PENDING] LOESS_filtered/ECO_L2T_LSTE.002_LST_doy2020063232055_aid0001_2S_LOESS_filtered.tif (EPSG:32702 to EPSG:4326)
[PENDING] LOESS_filtered/ECO_L2T_LSTE.002_LST_doy2020071201315_aid0001_2S_LOESS_filtered.tif (EPSG:32702 to EPSG:4326)
[PENDING] LOESS_filtered/ECO_L2T_LSTE.002_LST_doy2020075183922_aid0001_2S_LOESS_filtered.tif (EPSG:32702 to EPSG:4326)

SkipRun Reprojection

8 Verifying Results

Always inspect LOESS outputs before using them for downstream analysis.

- Use the baseline graph tab to confirm that individual pixel baselines and the averaged baseline look stable.
- Use the scene preview to compare the Original and LOESS filtered panels before applying the filter to all scenes.
- Click a few pixels in the preview to confirm displayed temperatures are reasonable.
- Load one original scene and its matching LOESS_filtered output into QGIS and toggle the layers on and off.
- Inspect cloud fringes, NaN boundaries, coastlines, and any unusual hot/cold areas.
- Open the saved parameters text file to confirm the final n_desired, depth_required, n_candidates_initial, frac_val, it_val, threshold_cold, threshold_hot, and seed.
- Load sample_pixels.shp in QGIS to verify which pixels were used to build the averaged baseline.
- If scenes were reprojected into EPSG:4326, verify their CRS and geolocation in QGIS.

9 Troubleshooting

Issue	Possible Cause	Suggested Fix
No LST TIFF scenes found	Filenames do not contain recognizable LST tokens, files are not .tif/.tiff, or files are inside excluded output folders.	Check filenames and selected root folder. Move raw scenes outside LOESS_filtered and LOESS_tuning_parameters folders.
Capacity limit exceeded	Too many scenes or very large inputs for the current workflow settings.	Split the dataset by year, season, tile, or study period and run the same tuning workflow on each subset.
LOESS build failed	Not enough valid candidate pixels, scenes do not overlap, or scenes do not share the same CRS.	Increase n_candidates_initial, lower depth_required, check that scenes overlap, and reproject inputs before running LOESS.
Baseline graphs look noisy	frac_val may be too low or candidate pixels may be unstable.	Increase frac_val or try a different sample seed.
Baseline graphs look too smooth	frac_val may be too high.	Lower frac_val and rebuild LOESS graphs.
Too many pixels removed	threshold_cold or threshold_hot may be too strict.	Increase the threshold values and preview again.
Cloud fringes remain	Thresholds may be too conservative or the baseline may not represent the dataset well.	Lower threshold_cold/threshold_hot gradually, adjust frac_val, or try a different seed.
sample_pixels.shp fails to save	Fiona is missing or not installed correctly.	Install Fiona with pip install fiona, then save the parameter snapshot again.
Reprojection popup finds no files	Outputs may already be EPSG:4326 or LOESS_filtered may be empty.	Check LOESS_filtered contents and CRS in QGIS.

10 Acknowledgements & Licensese

User Guide & Reference Manual developed by Jacquelyn Meiseles, NASA Jet Propulsion Laboratory, California Institute of Technology.

GUI developed by Jacquelyn Meiseles and Jake Longenecker with inspiration from GeoViewer (Longenecker, 2026).

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